

SETU-STAT

AI-Enabled Competency & Learning Platform
for India's Official Statistical System



Smart India Hackathon 2026 | PS ID: SIH26101 | Ministry of Statistics and Programme Implementation (MoSPI)

A Complete Solution Blueprint

From Problem Statement Analysis to Working Prototype

Problem Statement ID	SIH26101
Organization	Ministry of Statistics and Programme Implementation (MoSPI)
Category	Software – EdTech / GovTech / Artificial Intelligence
Theme	Smart Education / Capacity Building
Proposed Solution Name	SETU-STAT (Statistical Training & Evaluation Unified platform)
Tagline	Bridging the Competency Gap in India's Statistical System, One Learner at a Time

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1. Understanding the Problem Statement

The Ministry of Statistics and Programme Implementation (MoSPI) anchors India's Official Statistical System (OSS) – the network of central and state agencies that produce national accounts, price indices, labour force surveys, census-adjacent data and other official statistics that drive planning and policy. The people who run this system – Investigators, Statistical Officers, Assistant Directors, Deputy Directors and technical staff of the Indian Statistical Service (ISS) and subordinate cadres – must continuously upgrade skills in survey methodology, sampling theory, data science, official statistics standards (SDMX, NSSO schedules), and emerging tools like AI/ML and GIS-based statistics.

MoSPI already possesses **iGOT Karmayogi**, the Government of India's flagship integrated learning management system used across ministries for capacity building of civil servants (part of Mission Karmayogi / NPCSCB). However, iGOT is largely a *content delivery* platform – it does not natively know **what a specific officer is weak in**, cannot **auto-generate assessments** from new training material, and does not **personalize a learning path** based on measured competency gaps. The problem statement asks us to build the missing intelligence layer on top of / alongside this ecosystem.

Verbatim Problem Statement (Paraphrased)

“Develop an AI-enabled learning platform that identifies competency gaps, recommends personalized training through integration with the iGOT Karmayogi ecosystem, and is capable of generating quizzes and multiple-choice questions (MCQs) from uploaded learning materials to strengthen capacity building in India's Official Statistical System.”

Breaking Down the Ask – Three Explicit Deliverables

#	Explicit Requirement	What It Really Means
1	Identify competency gaps	Diagnose, per officer/role, which statistical & digital competencies are below the required proficiency level
2	Recommend personalized training via iGOT integration	Don't reinvent content – pull/push course recommendations from iGOT Karmayogi's catalogue via API/SSO
3	Auto-generate quizzes/MCQs from uploaded material	Any SME/trainer can upload a PDF/PPT and the system produces a valid, graded assessment instantly

Implicit (unstated but expected) requirements that separate a good submission from a winning one: role-based competency frameworks specific to the statistical cadre; scalability to lakhs of government employees across States/UTs; multilingual support (Hindi + regional languages); offline/low-bandwidth access for field-level Investigators; auditability and explainability of AI decisions (critical for government systems); data privacy/security compliance (CERT-In, GIGW); and measurable ROI for MoSPI leadership.

2. Deep-Dive: Requirements Analysis

2.1 Functional Requirements

- **FR1 – Competency Framework Engine:** Maintain a structured ontology of statistical competencies (sampling design, estimation theory, data visualization, MS-Excel/R/Python, SDMX reporting, ethics, etc.) mapped to cadre/role/grade.
- **FR2 – Skill Assessment:** Baseline diagnostic tests, self-assessment surveys, and passive signals (course history, on-the-job performance data where available) to build a Competency Vector per user.
- **FR3 – Gap Analysis Engine:** Compare current competency vector against a target profile (role-based) to compute a quantified, explainable Gap Score per skill.
- **FR4 – Recommendation Engine:** Suggest ranked iGOT Karmayogi / DIKSHA courses (and internal content) that close the highest-severity gaps first, considering learner preference and time budget.
- **FR5 – iGOT Karmayogi Integration:** SSO login, course-catalogue sync, enrolment push-back, completion & certificate sync (via iGOT's exposed APIs / SCORM or MoU-based data-sharing).
- **FR6 – Content Ingestion:** Accept PDFs, PPTs, DOCX, scanned notes and video transcripts as training material.
- **FR7 – AI Quiz/MCQ Generation:** Automatically produce MCQs (with distractors, correct answer, explanation and difficulty tag) from the uploaded material using GenAI + RAG, mapped to Bloom's taxonomy levels.
- **FR8 – Adaptive Assessment:** Deliver quizzes adaptively (harder/easier next question based on performance) using Item Response Theory (IRT).
- **FR9 – Analytics Dashboards:** Individual, team and organization-level dashboards for HoDs, DDG (Training) and MoSPI leadership.
- **FR10 – Certification & Nudges:** Auto-generate completion certificates/CPD credits and send gamified nudges/reminders.

2.2 Non-Functional Requirements

- **Scalability:** Must support 50,000+ concurrent users across NSO, CSO, NSSO field offices and State DES units.
- **Availability:** 99.5%+ uptime; must work in low-bandwidth zones (field investigators) via PWA + offline caching.
- **Security & Compliance:** Aadhaar/DigiLocker-linked SSO, AES-256 encryption, RBAC, adherence to MeitY GIGW 3.0 and CERT-In guidelines; data residency within India (empanelled cloud, e.g. MeghRaj/NIC).
- **Explainability:** Every AI-generated recommendation/question must show its reasoning/source – a hard requirement for government trust and grievance redressal.
- **Accessibility:** WCAG 2.1 AA compliant; Hindi + 8 regional languages via Bhashini API.
- **Interoperability:** Must expose/consume open standards – SCORM/xAPI for learning records, SDMX for statistical metadata, OpenAPI for integration.

3. Research – Why Existing Solutions Fall Short

Existing System	Strength	Gap it leaves open
iGOT Karmayogi	Massive content library, official SSO, Govt-wide adoption, FRAC integration	No automatic competency-gap diagnosis; recommends generically, not per measured gap
DIKSHA	Rich K-12/teacher content engine	Not built for adult/civil-service competency modelling or statistical domain
Generic LMS (Moodle/Coursera-for-Govt)	Course hosting, basic quizzes	Manual quiz creation; no AI gap detection; no native iGOT interoperability
Manual training-needs-assessment (current MoSPI practice)	Human judgement, contextual	Slow (annual cycle), subjective, not data-driven, cannot scale to lakhs of staff

The white space: nobody has combined (a) a statistics-cadre-specific competency ontology, (b) automatic AI-driven gap scoring, and (c) generative-AI content-to-assessment pipelines, wired directly into iGOT Karmayogi's existing rails. That combination is exactly what SETU-STAT delivers — and it is the core of our differentiation.

4. Our Solution: SETU-STAT Overview

SETU-STAT ("Setu" = bridge in Hindi/Sanskrit) is an AI-enabled competency intelligence layer that sits *on top of and alongside* iGOT Karmayogi — it does not replace iGOT, it makes iGOT smart for the statistical cadre. It continuously senses what each officer knows, compares it to what their role demands, prescribes the exact iGOT/DIKSHA courses to close that gap, and — uniquely — lets any trainer turn a raw PDF/PPT into a validated quiz in under two minutes.

4.1 The Three Pillars (Mapped 1:1 to the PS)

Pillar	Module Name	One-line description
Diagnose	Competency Gap Analyzer (CGA)	NLP + Skill-Graph engine that scores every officer against a role-based statistical-competency framework
Prescribe	Hybrid Recommendation Engine	Collaborative + content-based + knowledge-graph recommender that maps gaps directly to iGOT Karmayogi course IDs
Assess	GenAI Quiz/MCQ Generator	RAG + LLM pipeline that converts any uploaded material into Bloom's-taxonomy-tagged, adaptive MCQs in minutes

End-to-End Officer Journey on SETU-STAT

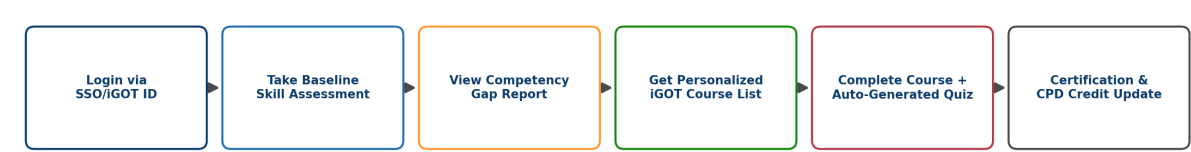


Figure 1: End-to-end journey of a statistical officer using SETU-STAT

5. What Makes SETU-STAT Uniquely Winning

- **Domain-first, not generic:** We don't ship a generic 'AI tutor'. We build a first-of-its-kind **Statistical Competency Ontology** aligned to NCF-SO (National Competency Framework for Statistical Officers, modelled after NCF/Frac used in Mission Karmayogi) – covering sampling theory, estimation, survey design, data ethics, GIS, R/Python for official statistics, and SDMX reporting.
- **Augments, doesn't replace, iGOT:** Judges consistently penalize solutions that try to 'rebuild' an existing government platform. SETU-STAT explicitly plugs into iGOT via its SSO, SCORM/xAPI and course-catalogue APIs — lower integration risk, faster real-world adoption, and directly aligned with 'Digital Public Infrastructure' (DPI) reuse philosophy.
- **Explainable-by-design AI:** Every recommendation shows 'why' (e.g., 'Recommended because your Sampling Design score is 42/100, below the 70 threshold required for Grade B Officers'). Every AI-generated MCQ cites the exact source paragraph/slide it was derived from — a hallucination-guard and audit trail that a government reviewer will trust.
- **Human-in-the-loop governance:** SME/trainer review workflow before high-stakes quizzes go live — balances speed of GenAI with the accountability government systems require.
- **Adaptive testing (IRT), not static quizzes:** Uses Item Response Theory so weaker learners aren't demoralized and strong learners aren't bored — rare in EdTech-for-government submissions.
- **Field-ready:** Progressive Web App with offline caching for Investigators in remote blocks with poor connectivity — directly addresses MoSPI's actual workforce distribution (most officers are field-based, not desk-based).
- **Multilingual by default:** Bhashini API integration for Hindi and regional-language quiz translation — critical since State DES (Directorate of Economics & Statistics) staff use vernacular languages.
- **Built for scale on Digital Public Infrastructure:** Uses DigiLocker/Aadhaar-linked SSO, cloud-native architecture on MeghRaj — shows judges we understand GovTech deployment constraints, not just a hackathon demo.
- **Closed feedback loop:** Quiz performance data flows back into the Competency Gap Analyzer — the system gets smarter about each officer over time (a true adaptive learning loop, not a one-shot recommendation).

6. System Architecture

The architecture is organized into six layers — from user access down to cross-cutting security/DevOps concerns — so each concern can be built, tested, and scaled independently. This is the reference architecture for both the hackathon prototype and the eventual production system (the prototype simply mocks/stubs the heavier integration and data-layer pieces).

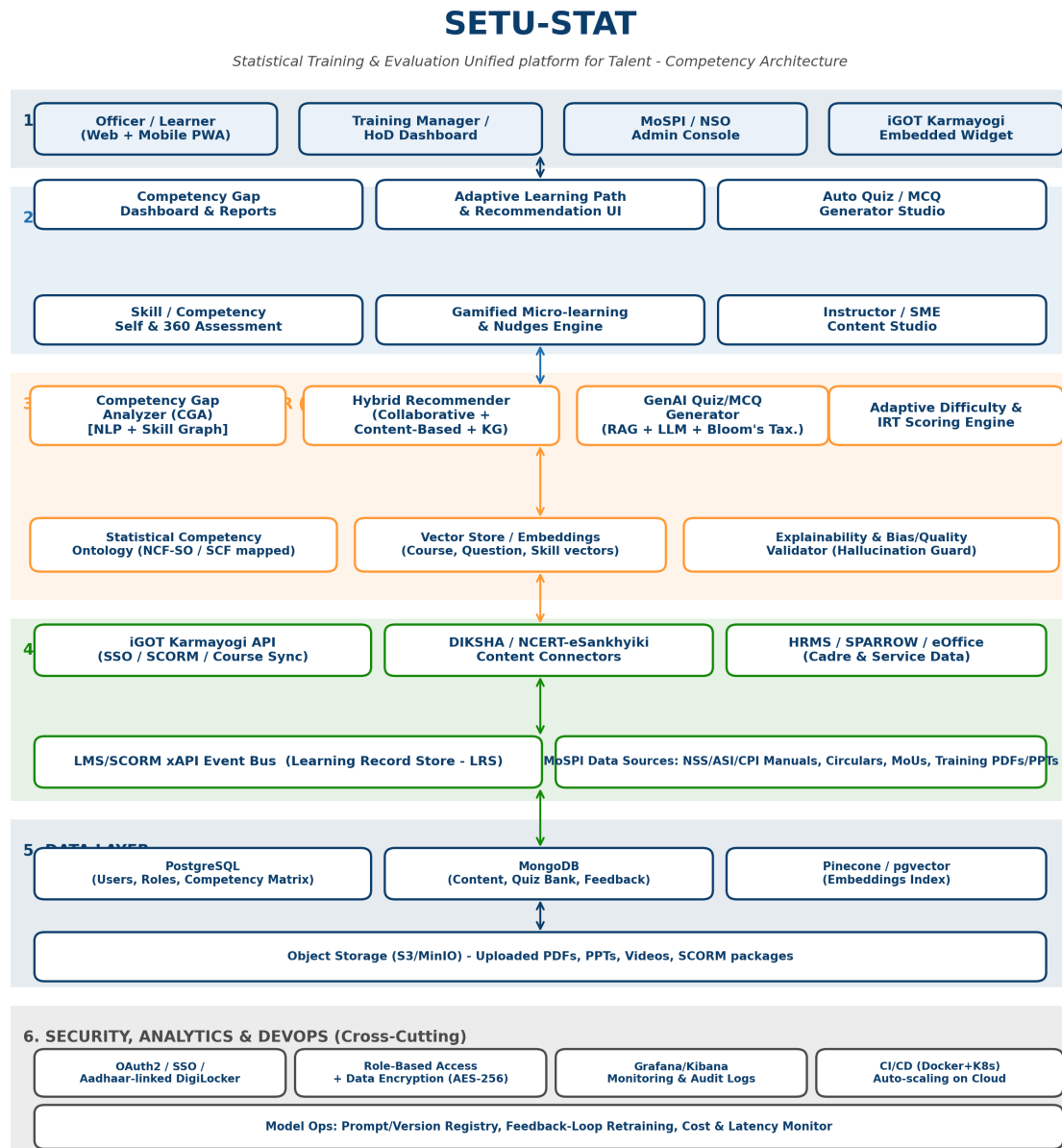


Figure 2: SETU-STAT six-layer reference architecture

7. Core Module 1: AI Competency Gap Analyzer (CGA)

The CGA is the diagnostic brain of the platform. It ingests structured and unstructured signals about an officer and produces a per-competency Gap Score (0–100) with full explainability.

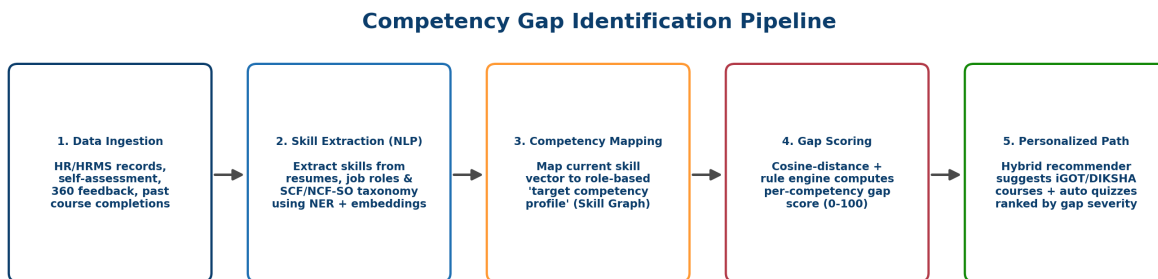


Figure 3: Five-stage Competency Gap Identification pipeline

7.1 How the Gap Score is Computed

Each officer is represented as a **Competency Vector** V_u across N statistical competencies (e.g., Sampling Design, Estimation Theory, Data Visualization, R/Python, Survey Ethics, SDMX/Metadata Standards, GIS for Statistics, Report Writing). A **Target Vector** T_r is defined per role/grade (e.g., Junior Statistical Officer vs Deputy Director) by an SME-curated competency-framework (digitized version of MoSPI's cadre training manuals + FRAC-aligned skill definitions). The Gap Score for competency i is:

$$\text{Gap}_i = \max(0, T_{r,i} - V_{u,i}) \times W_i$$

where W_i is a criticality weight (e.g., Survey Ethics and Sampling Design are weighted higher for field-facing roles). Signals feeding V_u include: baseline diagnostic test results, self-assessment surveys, iGOT course-completion history (via API), and (optionally, with consent) supervisor 360-feedback. NLP (Named Entity Recognition + sentence embeddings) is used to map free-text job descriptions/circulars to the standard competency taxonomy so the framework can evolve without manual re-tagging every time a new circular or scheme is introduced.

8. Core Module 2: Personalized Recommendation Engine

Once gaps are known, the Hybrid Recommender ranks the best available learning resources — prioritizing iGOT Karmayogi courses — to close the most severe/critical gaps first, respecting the officer's available time budget and preferred language.

8.1 Recommendation Strategy (Hybrid)

- **Content-Based Filtering:** Match competency-tagged course metadata (from iGOT's course catalogue API) against the officer's top-N gap competencies.
- **Collaborative Filtering:** 'Officers with a similar gap profile and role who completed Course X saw the fastest improvement' — learned from anonymized completion + post-test performance data.
- **Knowledge-Graph Reasoning:** A skill-prerequisite graph ensures foundational courses (e.g., 'Basics of Sampling') are recommended before advanced ones (e.g., 'Small Area Estimation'), preventing frustration.
- **Constraint Optimization:** A lightweight knapsack-style optimizer fits the recommended course bundle into the officer's declared weekly learning-time budget.

8.2 iGOT Karmayogi Integration Approach

Given iGOT Karmayogi is built on the Sunbird/DIKSHA open-source stack, integration is designed around its documented capabilities:

- **SSO (Single Sign-On):** Officers log into SETU-STAT using their existing iGOT/Karmayogi Bharat credentials (OAuth2/SAML), avoiding a new identity silo.
- **Course Catalogue Sync:** Periodic pull of course metadata (title, competency tags, duration, language) via iGOT's content-search API.
- **Enrolment Push-Back:** When SETU-STAT recommends a course, one click enrolls the officer directly on iGOT (deep link / enrolment API), so the actual learning experience stays on the platform officers already trust.
- **Completion & xAPI Sync:** Course completion, certificates and time-spent statistics flow back via SCORM/xAPI statements into SETU-STAT's Learning Record Store (LRS), closing the feedback loop into the Gap Analyzer.
- **Fallback for Hackathon Demo:** Since production iGOT API credentials are restricted to onboarded ministries, the prototype ships a realistic **Mock iGOT Connector** (same API contract, seeded with real publicly-listed iGOT course names) — so the integration code is production-ready on day one of real API access.

9. Core Module 3: GenAI Quiz / MCQ Generator

This module lets any trainer/SME upload raw learning material and receive a ready-to-deploy, quality-checked quiz — the single most time-consuming task in capacity-building today, reduced from hours to minutes.

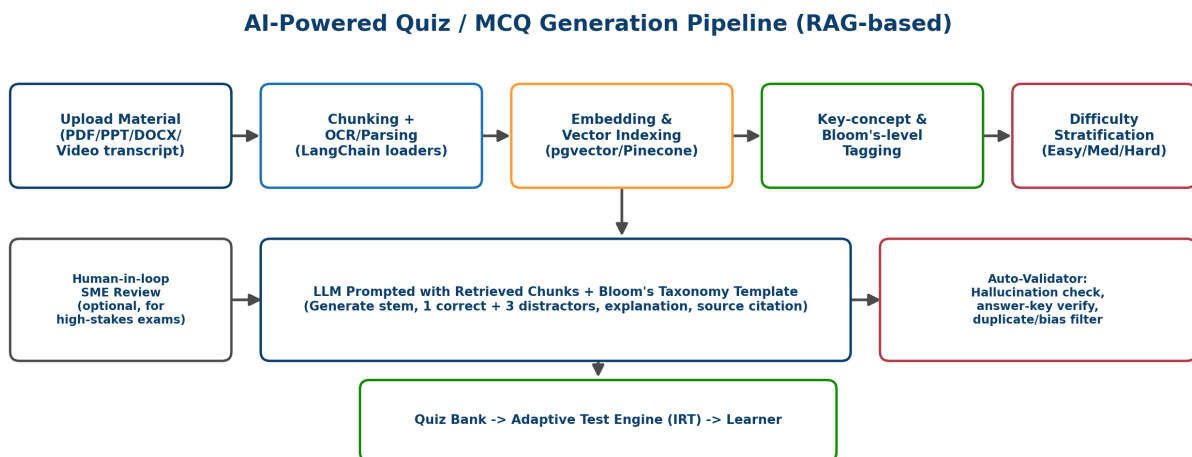


Figure 4: RAG-based Quiz/MCQ generation pipeline with human-in-the-loop validation

9.1 Prompt Engineering Approach (Simplified)

You are a subject-matter question-setter for India's Official Statistical System. Using ONLY the context below, write one multiple-choice question at Bloom's level: {level}. Return JSON with fields: question, options[4], correct_index, explanation, source_snippet. Do not introduce facts not present in the context. Context: {retrieved_chunks}

9.2 Quality & Trust Safeguards

- **Grounded generation (RAG):** The LLM only sees retrieved chunks from the uploaded document — not open-ended knowledge — minimizing hallucination.
- **Auto-Validator:** Rule-based + LLM-as-judge checks for duplicate questions, answer-key correctness, distractor plausibility, and biased/ambiguous phrasing before publishing.
- **Source citation:** Every question stores the exact page/slide number it was derived from, so a reviewer can verify in seconds.
- **Bloom's Taxonomy tagging:** Questions are auto-tagged Remember / Understand / Apply / Analyze so exams can be balanced across cognitive levels — a NCERT/UPSC-grade quality bar.
- **Adaptive delivery (IRT):** Once quizzes are live, an Item Response Theory engine selects the next question's difficulty based on the learner's running ability estimate θ , giving fairer, shorter, more diagnostic tests.

10. Technology Stack

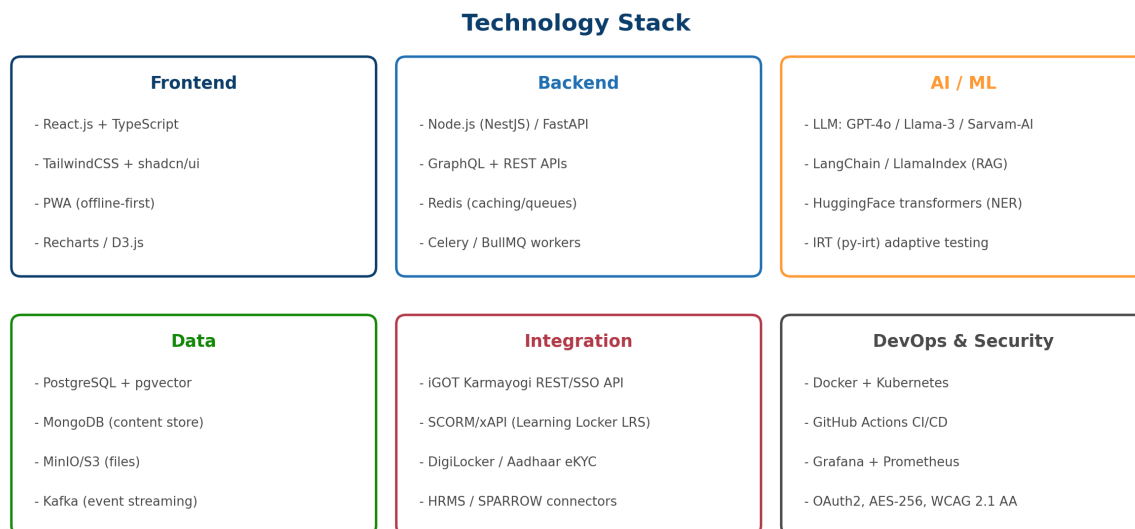


Figure 5: Recommended technology stack across layers

10.1 Why These Choices

- **FastAPI/NestJS** chosen for async performance under concurrent quiz-taking load and native OpenAPI docs (useful for judges/integration reviewers).
- **LangChain/LlamaIndex** for RAG orchestration — mature ecosystem, swappable LLM backends (can run GPT-4o for demo, switch to a sovereign/on-prem model like Sarvam-AI or fine-tuned Llama-3 for production data-residency compliance).
- **pgvector** (Postgres extension) keeps the prototype simple — one database for relational + vector data — while remaining swappable for Pinecone/Milvus at scale.
- **py-irt** for adaptive testing gives statistically rigorous ability estimation — fitting, since MoSPI is a statistics organization, using real Item Response Theory rather than ad-hoc heuristics is itself a credibility signal to judges.
- **Docker + Kubernetes** ensures the same containers that run in the demo can be deployed on MeghRaj (NIC's Government Community Cloud) with minimal change.

11. Data Flow & Security

11.1 Data Flow Summary

Uploaded material and officer interactions flow through: **Ingestion** → **Processing/AI** → **Storage** → **Serving** → **Feedback loop**. Personally identifiable information (name, employee ID, contact) is stored separately from behavioural/competency data with tokenized joins, so analytics pipelines and any external AI-model calls never see raw PII.

11.2 Security & Compliance Checklist

- OAuth2/SAML SSO via iGOT / DigiLocker — no parallel password store.
- Role-Based Access Control (Officer / Trainer / HoD / MoSPI Admin) with principle of least privilege.
- AES-256 encryption at rest, TLS 1.3 in transit.
- PII tokenization + separate encrypted vault; AI/LLM calls receive only anonymized competency vectors and document text, never personal identifiers.
- Audit logging of every AI-generated recommendation and quiz for grievance redressal (explainable-AI requirement for government systems).
- Deployment on empanelled Government Cloud (MeghRaj) / data residency within India; compliance with CERT-In and MeitY GIGW 3.0 guidelines.
- Periodic bias audit of the recommendation and quiz-generation models (fairness across gender, region, cadre).

12. Impact Assessment

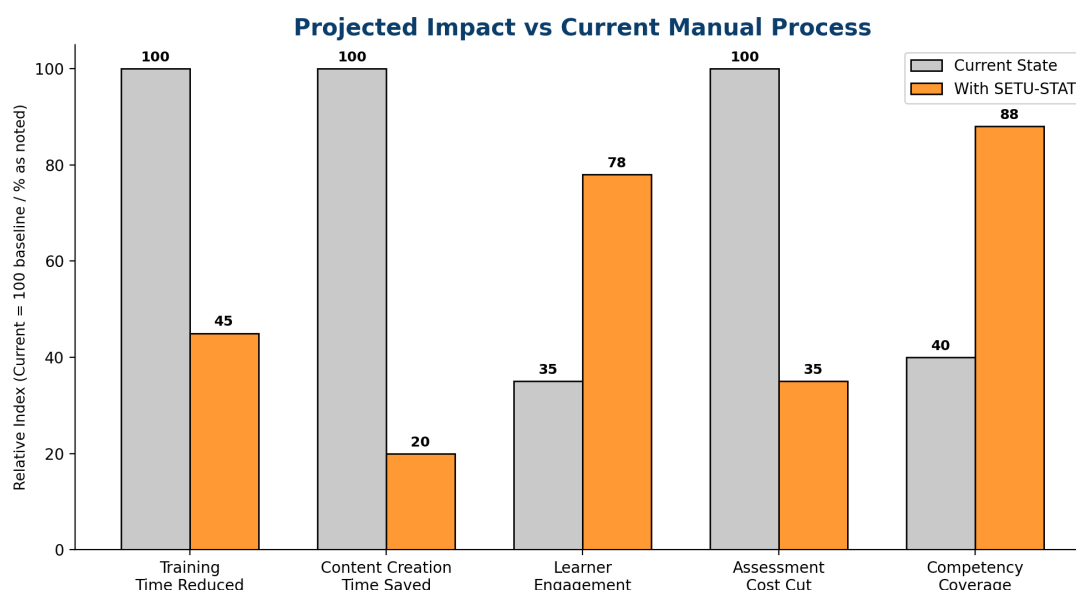


Figure 6: Projected impact vs current manual training-needs-assessment process (illustrative, to be validated with MoSPI pilot data)

12.1 Quantitative Impact (Illustrative Targets)

Metric	Current State	With SETU-STAT (Target)
Time to create one 20-question quiz	3–5 hours (manual)	< 10 minutes (AI-assisted + SME review)
Training-needs assessment cycle	Annual, survey-based	Continuous, real-time competency dashboard
Course-to-officer relevance	Generic, catalogue browsing	Personalized, gap-severity ranked
Officers reached per training cycle	Limited by trainer bandwidth	Scales to entire cadre (lakhs) simultaneously
Assessment cost per officer	High (proctor, printing, logistics)	Near-zero marginal cost (digital, adaptive)

12.2 Qualitative & Strategic Impact

- **Direct alignment with Mission Karmayogi's** vision of competency-based HR management across the entire civil service — SETU-STAT can act as a pilot 'vertical' that other ministries replicate.
- **Better statistics for India:** A more competent statistical workforce directly improves the quality of national data (GDP estimates, CPI, employment surveys) that drives every policy decision.
- **Equity across geographies:** Officers in remote/State DES offices get the same quality of personalized guidance as those in Delhi HQ.
- **Institutional memory:** As trainers upload materials over years, SETU-STAT becomes a living, searchable, continuously-assessed knowledge base of India's statistical practice.
- **Data-driven HR decisions:** Aggregated (anonymized) competency dashboards help MoSPI leadership plan recruitment, deputation and promotion criteria objectively.

13. Feasibility & Viability

Dimension	Assessment
Technical Feasibility	High — all components (RAG, IRT, recommender systems, SSO) are mature, well-documented technologies with strong open-source support.
Integration Feasibility	Medium-High — iGOT Karmayogi (Sunbird-based) publishes integration patterns for ministry-specific portals; formal API access requires MoSPI-NIC-DoPT coordination, which is a process risk, not a technical one.
Financial Feasibility	High — can be built on open-source stack + empanelled GCC cloud; LLM inference cost is the main recurring cost, mitigated by caching, batching and eventually a fine-tuned smaller model.
Adoption Feasibility	High — zero new login (SSO reuse), officers keep using iGOT for actual course consumption; SETU-STAT is additive, not disruptive.
Risks	LLM hallucination in quiz generation (mitigated by RAG + validator + human review); data-privacy sensitivities around performance data (mitigated by anonymization + RBAC); dependency on timely iGOT API access (mitigated by mock-connector-first design).

14. Working Prototype – Step-by-Step Build Guide

This section is a practical, hour-by-hour implementable guide for a 4–6 member team to build a convincing, judge-ready working prototype (not just a mockup) within a hackathon timeframe.

14.1 Step 1 – Project Scaffolding (Hours 0–2)

- Create a monorepo: /frontend (React+Vite+TS+Tailwind), /backend (FastAPI), /ai-services (Python microservice for RAG/LLM), /infra (docker-compose).
- Set up PostgreSQL (with pgvector extension enabled) and seed schema: users, roles, competencies, competency_scores, courses (mocked iGOT catalogue), quizzes, questions, attempts.
- Set up basic OAuth2/JWT auth scaffolding (mock 'Login with iGOT' button that simulates SSO for the demo).
- Design 4–5 Figma wireframes: Login, Competency Dashboard, Course Recommendations, Quiz Upload/Generation Studio, Adaptive Quiz-Taking screen.

14.2 Step 2 – Competency Gap Engine (Hours 2–8)

- Hard-code a starter Competency Ontology as a JSON/YAML file: ~15–20 statistical competencies grouped under Technical, Analytical, Domain and Digital skill clusters, each with 3 target proficiency levels (Junior/Mid/Senior officer).
- Build a simple 15–20 question baseline diagnostic (multiple-choice + self-rated Likert) that produces a raw Competency Vector on submission.
- Implement the Gap Score formula (Section 7.1) as a FastAPI endpoint: POST /api/gap-analysis → returns per-competency gap scores + overall readiness percentage.
- Visualize results as a radar/bar chart on the Competency Dashboard (Recharts).

14.3 Step 3 – Recommendation Engine + Mock iGOT Connector (Hours 8–14)

- Seed a courses table with 30–50 real/plausible iGOT-style course titles tagged to competencies (this is your 'Mock iGOT Karmayogi Connector' — built to the same API contract iGOT would expose, so it's a drop-in replacement later).
- Implement content-based filtering: cosine similarity between the officer's gap vector and each course's competency-tag vector; sort descending.
- Add a simple prerequisite rule (knowledge-graph-lite): don't recommend 'Advanced Sampling' before 'Basics of Sampling' is marked complete.
- Build the 'Recommended for You' UI card list with a one-click (mock) 'Enrol on iGOT' deep-link button.

14.4 Step 4 – GenAI Quiz/MCQ Generator (Hours 14–24 – the core wow-factor)

- Build a file-upload endpoint accepting PDF/PPTX; use pypdf/python-pptx to extract text, chunk into ~500-token passages.
- Embed chunks using a sentence-transformer (or OpenAI/embedding API) and store in pgvector.
- For quiz generation, retrieve top-k relevant chunks per requested topic, and call an LLM (OpenAI/Anthropic/open-source) with a strict JSON-output prompt (Section 9.1) requesting stem, 4 options, correct index, explanation and source snippet.

- Parse and validate the JSON; run a lightweight auto-check (no duplicate correct-answer patterns like always 'C', options roughly equal length, explanation non-empty).
- Render an editable review screen (SME can tweak before publishing) — this is your human-in-the-loop safeguard, and a great live-demo moment.
- Store finalized questions in the quiz bank tagged with difficulty (ask the LLM to self-rate, or infer from readability/complexity heuristics).

14.5 Step 5 – Adaptive Quiz Delivery (Hours 24–30)

- Implement a simplified 1-parameter IRT (Rasch) model using the `py-irt` or a hand-rolled logistic-difficulty function: after each answer, update the learner's ability estimate θ and pick the next question whose difficulty is closest to θ .
- Show a live 'ability meter' during the quiz for a strong visual demo effect.
- On completion, push the score back into the Competency Gap Engine to auto-update that competency's score (closing the feedback loop) and re-render the dashboard.

14.6 Step 6 – Polish, Sample Data & Demo Script (Hours 30–36)

- Seed 3–4 realistic demo personas (e.g., 'Junior Statistical Officer, Bihar DES' with a deliberately weak Sampling Design score) so the gap-to-recommendation-to-quiz story is visually compelling.
- Add empty/loading/error states so the UI never looks broken mid-demo.
- Record a 2–3 minute backup video walkthrough in case of live-demo connectivity issues (a common SIH judging-round risk).
- Prepare the pitch deck mapping every screen back to a Problem Statement requirement (judges specifically reward clear PS-traceability).

15. 36-Hour Hackathon Roadmap

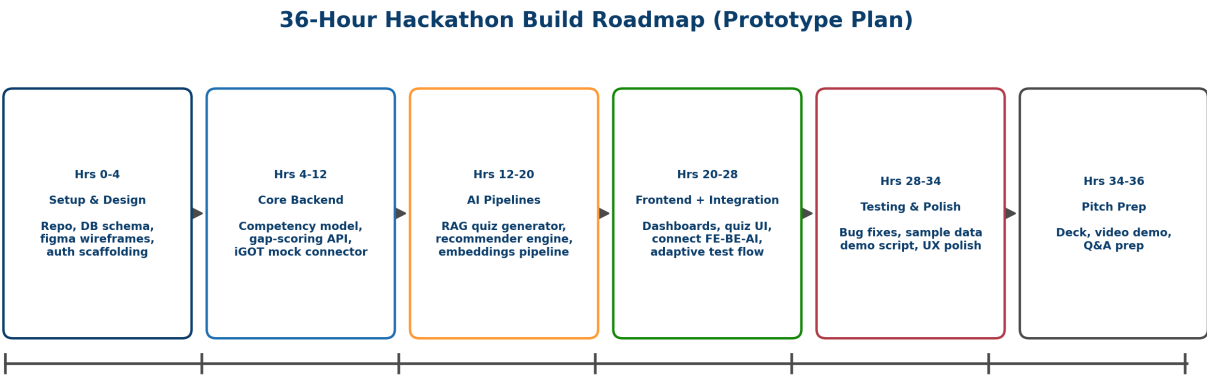


Figure 7: Suggested build roadmap and time allocation

15.1 Suggested Team Split (5 members)

Role	Owns	Key Hours
Backend/API Lead	FastAPI services, DB schema, gap-score API, auth	0–30
AI/ML Engineer	RAG pipeline, embeddings, LLM prompt design, IRT model	2–30
Frontend Lead	React dashboards, quiz UI, upload studio	0–32
Integration/DevOps	Mock iGOT connector, Docker, deployment, demo data seeding	4–34
Presentation/Research Lead	Ontology curation, pitch deck, demo script, judge Q&A; prep	0–36

16. Demo Script for Judges (5-minute flow)

- **[0:00–0:30] Hook:** "India's official statistics — GDP, CPI, unemployment — are only as good as the officers who produce them. Today, identifying who needs what training is a manual, annual, subjective process. We built SETU-STAT to fix that."
- **[0:30–1:30] Show the Gap:** Log in as 'Junior Statistical Officer – Bihar DES'. Show the Competency Dashboard — point to the lowest-scoring competency (e.g., Sampling Design: 42/100) and explain how that score was computed.
- **[1:30–2:30] Show the Recommendation:** Scroll to 'Recommended for You' — show 2–3 ranked iGOT courses explicitly tied to the weak competency, with the explainability tooltip ('Recommended because...'). Click 'Enrol on iGOT' deep link.
- **[2:30–4:00] Show the Wow-Factor – Live Quiz Generation:** Upload a real MoSPI-style training PDF (e.g., a public NSSO survey methodology document) live. Within seconds, show 5 generated MCQs with source citations. Open the SME review screen, tweak one distractor, publish.
- **[4:00–4:40] Show Adaptive Testing:** Take 3–4 questions of the quiz live, showing the ability meter adjusting difficulty in real time.
- **[4:40–5:00] Close on Impact & Scale:** Return to the impact chart — reiterate that this plugs into infrastructure MoSPI already has (iGOT), needs no new officer logins, and is ready to pilot with a single State DES office.

17. Business Model & Sustainability (Post-Hackathon)

- **Phase 1 – Pilot (0–6 months):** Deploy with one willing State DES / NSSO regional office; validate gap-scoring accuracy against actual performance/appraisal outcomes.
- **Phase 2 – iGOT Native Integration (6–12 months):** Formal API onboarding with NIC/Karmayogi Bharat team; SETU-STAT ships as an official 'Competency Intelligence' plug-in inside iGOT for the statistics vertical.
- **Phase 3 – Cross-Ministry Replication (12–24 months):** The competency-ontology + gap-engine architecture is domain-agnostic — the same platform (with a different ontology) can serve any Central/State department under Mission Karmayogi, creating a natural expansion path and shared infrastructure cost.
- **Funding Path:** Government budget line under Digital India / Mission Karmayogi capacity-building allocation, or empanelment as a GeM (Government e-Marketplace) SaaS offering for reuse by other ministries.

18. Future Roadmap

- **Voice-based micro-learning** for field Investigators via IVR/WhatsApp integration for areas with no smartphone/data access.
- **Predictive attrition/performance analytics:** Flag officers likely to need urgent upskilling before it affects survey data quality.
- **Fine-tuned domain LLM:** Train a smaller, on-prem model specifically on MoSPI's own manuals/circulars to reduce cost and improve data sovereignty over time.
- **Peer-learning & mentorship matching:** Connect officers with a gap in Skill X to a high-scoring peer for informal mentorship.
- **Integration with performance appraisal (APAR)** systems for a fully closed-loop, evidence-based HR capability system across the statistical cadre.

19. Team Roles & Skill Requirements

Skill Area	Tools/Frameworks	Why Needed
Full-stack Web Development	React, TypeScript, FastAPI/Node	Dashboards, upload studio, quiz UI
Applied ML / NLP	HuggingFace, sentence-transformers, scikit-learn	Skill extraction, embeddings, recommender logic
GenAI / Prompt Engineering	LangChain/LlamaIndex, OpenAI/Anthropic/local LLM APIs	RAG-based quiz generation, explanation generation
Data Engineering	PostgreSQL, pgvector, Docker	Schema design, vector search, containerization
UX/Product Design	Figma, accessibility (WCAG) knowledge	Officer-friendly, low-literacy-tolerant UI
Domain/Research	Understanding of MoSPI, NSSO, iGOT/Karmayogi ecosystem	Grounding the ontology & pitch in real government context

20. Conclusion

SETU-STAT directly answers all three explicit asks of SIH26101 — competency-gap identification, iGOT-integrated personalized recommendations, and AI-generated quizzes/MCQs — while layering in the details that distinguish a good hackathon demo from a genuinely deployable government solution: an explainable, statistics-specific competency ontology; respectful, additive integration with existing Digital Public Infrastructure (iGOT Karmayogi) rather than a competing silo; human-in-the-loop governance over generative AI; and a field-ready, multilingual, low-bandwidth-tolerant design built for India's real statistical workforce. The result is a platform that doesn't just win a hackathon demo — it maps to a credible, fundable, 12–24-month path to actually strengthening the backbone of India's official statistics.

SETU-STAT — Bridging the Competency Gap in India's Statistical System, One Learner at a Time.